The Joint Episode as an Ontological-Temporal Unit:

Mathematical Characterization, Structural Properties, and Transitions in Complex Systems

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Abstract

We develop the Joint Episode as a mathematically precise and ontologically primitive temporal unit for the study of complex systems. Grounded in the three-layer framework of systemic persistence, a Joint Episode is defined as a maximal interval of sustained low antisynchrony $A(k)$ (Layer 2: scale coherence and permeability) that contains at least one significant excursion of the critical-mass metric M (Layer 1: local intensification of Systemic Tau τ_s combined with RQA trapping).

We introduce the core quantitative descriptors — Joint Score $J = D \cdot \overline{M}$, duration D, intensity profile, and a multi-feature calibrated Confidence Score — and derive their structural relations to anti-synchronization, active memory, and layer-transition probabilities. Controlled experiments on coupled logistic maps and the Lorenz attractor demonstrate regime-dependent statistics of Joint Episodes and clear separation between episodes that precede detectable structural reorganization (Layer 3) and those that do not.

Ontologically, the Joint Episode realizes a coherent *kairos*: a structured window of the present in which intensified persistence acquires the relational conditions necessary for global reorganization. The framework thereby supplies both an operational detection algorithm and a non-reductive account of how local becoming participates in systemic transitions.

Keywords: Joint Episode; ontological-temporal unit; Systemic Tau; recurrence quantification analysis; antisynchrony; complex systems; structural transitions; persistence; kairos.

1 Introduction

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The analysis of complex systems has long sought intermediate constructs that are simultaneously mathematically tractable and ontologically meaningful. Recurrence-based methods provide powerful descriptors of local persistence and determinism, yet translating these descriptors into coherent units of analysis that respect both the mathematics of nonlinear dynamics and the structure of temporal becoming remains an open challenge.

Within the RECD + Ontología del Presente program, the three-layer framework distinguishes (i) local intensification of persistence (Layer 1), (ii) sustained coherence of persistence scales across modules (Layer 2), and (iii) global structural reorganization (Layer 3). The central proposal of this paper is that the **Joint Episode** — the co-occurrence of Layer 1 within a sustained Layer 2 window — constitutes a primitive ontological-temporal unit.

This paper develops the Joint Episode rigorously on three fronts:

- Formal mathematical definition and associated metrics (Joint Score, Confidence Score, duration, intensity).
- Derivation of structural properties and relations to anti-synchronization, memory, and layer transitions.
- Validation in controlled synthetic systems (coupled logistic maps, Lorenz attractor) together with an explicit ontological interpretation.

The emphasis is technical and mathematical, with the ontological dimension articulated primarily in its dedicated section while informing the formal development throughout.

3 Formal Definition of the Joint Episode

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4.1 Background: Systemic Tau and Antisynchrony

Systemic Tau $\tau_s(k)$ is a local measure of ordinal persistence derived from permutation entropy over sliding windows. Antisynchrony is defined as

$$A(k) = \mathrm{std}(\{\tau_s^i(k)\}_{i=1}^N),$$

where the standard deviation is taken across the N modules (or regions) at time k. Low sustained values of $A(k)$ indicate alignment of persistence scales (Layer 2).

4.2 Layer 1 Intensification

Layer 1 is identified by excursions of the critical mass metric $M(k)$, which combines normalized hyper-persistence of τ_s with RQA-derived trapping time (or laminarity) under Config B parameters.

4.3 Definition of the Joint Episode

Definition: Joint Episode

A **Joint Episode** is a maximal contiguous interval $[t_{\text{start}}, t_{\text{end}}]$ such that:

1. $A(k) < \theta_A$ for all $k \in [t_{\text{start}}, t_{\text{end}}]$ and $t_{\text{end}} - t_{\text{start}} + 1 \geq D_{\min}$ (sustained Layer 2),
2. there exists at least one $k^* \in [t_{\text{start}}, t_{\text{end}}]$ with $M(k^*) \geq \theta_M$ (presence of Layer 1 intensification).

The interval is maximal: it cannot be extended without violating either condition.

The thresholds θ_A, $D_{\min}$, and θ_M are regime-calibrated (see Section 3).

5 Metrics and Characterization

6 Metrics and Characterization

6.1 Primary Episode-Level Metrics

- **Duration** $D = t_{\text{end}} - t_{\text{start}} + 1$.
- **Joint Score** $J = D \times \overline{M}$, where $\overline{M}$ is the mean of $M(k)$ over the episode (alternative: integral or max-M variants).
- **Intensity profile**: mean M, max M, fraction of episode above threshold.
- **Layer 1 consistency**: number and clustering of high-M excursions within the episode.

6.2 Confidence Score

The Confidence Score is a weighted aggregate of normalized episode features (duration, Joint Score, Layer 1 consistency, proximity to Layer 3 indicators when available). It is calibrated on synthetic ensembles and yields categorical levels (e.g., Bajo / Medio / Alto).

6.3 Relation to Active Memory and Anti-synchronization

Joint Episodes quantify intervals during which active memory (persistence) can propagate because fragmentation (high $A(k)$) is suppressed.

7 Structural Properties

8 Structural Properties

8.1 Relation to Bidirectional Filtering

Layer 2 (low $A(k)$) acts as an ascending filter: only Layer 1 events occurring inside it have substantial probability of contributing to Layer 3. Layer 3 events exert descending constraints on acceptable Layer 1–2 configurations.

8.2 Properties of the Joint Score

- Monotonicity with persistence intensification and duration. - Regime dependence: distribution of J differs markedly between high-permeability (frequent long episodes) and low-permeability regimes. - Sensitivity to coupling strength in synthetic maps.

8.3 Active Memory Interpretation

Within a Joint Episode the system maintains a coherent "present" across scales long enough for local intensifications to become structurally consequential.

9 Transitions and Layer Dynamics

10 Transitions and Layer Dynamics

10.1 Probabilistic Transition Model

The probability of observing a Layer 3 event conditional on a Joint Episode of score J is modeled as an increasing function $P(L3 \mid J)$ subject to stochastic modulation. Empirical and synthetic data show clear separation in J distributions between episodes that precede detectable reorganization and those that do not.

10.2 Temporal Ordering

The canonical sequence is Layer 2 window formation $\rightarrow$ Layer 1 intensification(s) inside the window $\rightarrow$ elevated probability of Layer 3. Deviations and the conditions under which Layer 1 occurs outside Layer 2 (fragmented) are examined.

11 Synthetic Validation

11.1 Motivation and Scope

A central claim of the three-layer ontology developed in this work is that the **Joint Episode** constitutes a primitive ontological-temporal unit situated at the interface between local intensification of persistence (Layer 1) and relational coherence across scales (Layer 2). A natural and empirically testable implication of this claim is that such episodes should carry anticipatory information about subsequent structural reorganization (Layer 3).

To evaluate this implication under controlled conditions, we constructed an iterative diagnostic pipeline on synthetic dynamical systems. The design deliberately separates two distinct questions: (i) whether controlled structural perturbations produce a detectable signature in the Layer-2 observable $A(k)$, and (ii) to what extent the scalar scores $J = D \cdot \overline{M}$ and $J_{0.7} = D^{0.7} \cdot \overline{M}$ can discriminate episodes that precede genuine Layer-3 events from those that do not.

Crucially, we treat the second question not as a prediction task to be optimized, but as a **diagnostic probe** of the alignment between the proposed temporal unit and the underlying ontological layers. All experiments were performed with identical detection parameters, post-episode windows, and evaluation metrics across labeling regimes, ensuring direct comparability.

11.2 Synthetic Generators and Experimental Design

Two minimal modular systems were examined: four coupled logistic maps ($r = 3.8$) and four diffusively coupled Lorenz oscillators. The former served as an initial testbed; the latter was adopted after systematic diagnostic failure of the logistic generator to produce a reliable Layer-3 signature.

Coupled logistic maps (v0.1–v0.3). Structural perturbations were implemented as temporary, localized reductions of effective coupling (and optionally of the local growth parameter) applied to one or two modules. Despite aggressive schedules—including near-total decoupling over intervals of 70–110 steps—the mean post-perturbation shift in antisynchrony remained negative or statistically indistinguishable from matched random controls ($\Delta A \approx -0.011$ to -0.020). Aligned trace analysis revealed that any post-episode rise in $A(k)$ was dominated by generic rebound dynamics following low-antisynchrony intervals rather than by the injected structural events. This generator was therefore abandoned for the main validation.

Coupled Lorenz oscillators (v0.4 onward). The primary generator was replaced by four Lorenz oscillators integrated via classical RK4 ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) with mean-field diffusive coupling on the x-component. Structural perturbations consisted of temporary local reductions of coupling strength (down to $\varepsilon \approx 0.15$) and, optionally, local scaling of ρ (factors 0.70–0.92) for intervals of 60–100 steps. Under this dynamics, the first reliably positive mean post-perturbation shift in antisynchrony was observed: $\Delta A = +0.0249$ versus $+0.0025$ for temporally matched random controls (aggregated over 60 realizations). All subsequent labeling and scoring experiments were performed exclusively on this generator while preserving identical meta-parameters ($D_{\min} = 7$, post-window $[50, 150]$ in metric time, quantile-calibrated thresholds).

11.3 Joint Episode Detection and Scoring

Joint Episodes were detected exactly as defined in Section 4: maximal contiguous intervals during which $A(k) < \theta_A$ for duration $D \geq D_{\min}$ that also contain at least one excursion with $M(k) \geq \theta_M$. Thresholds were calibrated per realization on the empirical quantiles of the observed series. Two scalar scores were attached to every detected episode:

$$J = D \cdot \overline{M}, \qquad J_{0.7} = D^{0.7} \cdot \overline{M}.$$

The exponent 0.7 was introduced to attenuate linear dependence on duration while retaining sensitivity to intensity. All downstream discrimination statistics were computed on both scores using identical episode sets.

11.4 Layer-3 Labeling Regimes

Two labeling regimes were contrasted on the Lorenz generator.

Enriched hybrid labeling (v0.5). A scalar Layer-3 signal was constructed as a weighted combination of five features designed to capture both the magnitude and sustainability of post-episode reorganization:

$$l_3\text{-signal} = 0.22 \cdot \mathrm{frob}_x + 0.18 \cdot \mathrm{frob}_\tau + 0.28 \cdot \mathrm{ReLU}(\Delta A) + 0.18 \cdot \mathrm{persist} + 0.14 \cdot |\Delta \overline{\tau}|.$$

The threshold was set at the 0.84 quantile of the same signal evaluated on 250 random candidate times (independent of detected episodes). This regime produced labels with significantly improved ontological grounding: episodes labeled $Y = 1$ exhibited markedly stronger and more sustained post-episode elevation of $A(k)$ than those labeled $Y = 0$ (Mann–Whitney $p = 0.0066$).

Pure oracle labeling (v0.6, diagnostic upper bound). A separate diagnostic mode was implemented that completely bypasses the hybrid signal. An episode receives label $Y = 1$ if and only if at least one ground-truth structural perturbation onset falls within the identical post-episode window $[t_e + 50, t_e + 150]$ (metric time). This mode supplies a theoretical ceiling: the highest AUC that J or $J_{0.7}$ could possibly achieve under perfect knowledge of which episodes precede actual structural events. Oracle labeling is used exclusively for diagnostic purposes.

11.5 Results

Table 1 summarizes the evolution of discrimination performance across iterations. All Lorenz figures are based on 60 realizations ($T = 2000$, 4 modules).

Table 1: Evolution of discrimination performance across synthetic-validation iterations. Spearman correlations are reported for the pooled episode population.

Iteration	Generator	Labeling	AUC_J	$\mathrm{AUC}_{J_{0.7}}$	Spearman ρ (J / $J_{0.7}$)
v0.1	logistic	basic hybrid	0.494	0.506	0.79 / 0.64
v0.2	logistic	improved hybrid	0.501	0.512	0.78 / 0.63
v0.3	logistic	strengthened	0.471	0.474	0.68 / 0.51
v0.4	Lorenz	hybrid	0.503	0.476	0.75 / 0.61
v0.5	Lorenz	enriched hybrid	0.481	0.458	0.75 / 0.61
v0.6	Lorenz	oracle (ground truth)	0.469	**0.476**	0.75 / 0.61

The transition from logistic to Lorenz (v0.3 → v0.4) marks the first regime in which structural perturbations produce a reliably positive mean shift in antisynchrony. However, the decisive diagnostic result appears in v0.6. Even under **perfect ground-truth labeling** of episodes that precede actual structural perturbations, the Joint Scores achieve only modest discrimination ($\mathrm{AUC}_{J_{0.7}} \approx 0.476$). Moreover, episodes labeled positive by the oracle tend to exhibit *slightly lower* scores than negative episodes, and post-episode ΔA is not significantly different between the two classes (Mann–Whitney $p = 0.71$).

Importantly, the bias-reduction property of $J_{0.7}$ remains fully intact across all labeling regimes.

11.6 Discussion and Ontological Implications

The synthetic validation yields three principal findings.

First, the logistic generator proved fundamentally inadequate for testing the Layer-3 implication of the ontology. Even under aggressive structural perturbations, the observable $A(k)$ failed to register a consistent reorganization signature. This limitation is informative: localized coupling changes are rapidly washed out in strongly chaotic, low-dimensional mean-field systems with sliding ordinal statistics.

Second, the coupled Lorenz system succeeds in producing a detectable Layer-3 signature ($\Delta A = +0.0249$). The enriched hybrid labeling regime (v0.5) further succeeded in aligning binary labels Y with episodes followed by stronger and more sustained reorganization.

Third—and most consequential—the oracle upper-bound experiment (v0.6) reveals that even with perfect knowledge of which Joint Episodes precede ground-truth structural events, the scores J and $J_{0.7}$ possess only limited discriminative power. This indicates that the information captured by duration and mean intensity of high-M excursions within low-$A(k)$ windows is primarily diagnostic of **Layer 1–2 coherence** rather than strongly anticipatory of Layer-3 reorganization, at least in minimal four-dimensional coupled systems.

These findings support the Joint Episode as a well-defined ontological-temporal unit realizing coherent persistence across scales (Layers 1 and 2), while revealing that its capacity to anticipate structural transitions (Layer 3) is modest in low-dimensional settings. Stronger anticipatory power will likely require higher-dimensional or hierarchically structured generators, richer observables, or empirical systems with slower Layer-3 drivers.

11.7 Summary

Controlled synthetic experiments demonstrate that structural perturbations can induce detectable reorganization in $A(k)$ when the generator permits it, that enriched labeling improves ontological grounding of Layer-3 labels, and that even under oracle labeling the proposed Joint Scores achieve only modest discrimination of episodes that precede reorganization. The bias-reduction property of $J_{0.7}$ is robust across labeling regimes. These results delimit both the strengths and current limitations of the framework.

12 Ontological Implications

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13.1 The Joint Episode as Kairos

In the metaphysics of becoming, the present is not a point but a structured window of persistence. The Joint Episode realizes a coherent, detectable *kairos*: a temporal interval whose internal organization (persistence intensification + scale alignment) materially raises the likelihood of qualitative change.

13.2 Act of Being and Futurization

Drawing on the three-layer framework, the Joint Episode is the locus where the act of being (persistence) is both locally intensified and relationally coherent, thereby opening the possibility of "futurizing the present" — allowing local structure to participate in global reorganization.

13.3 Non-Reductive Hierarchy

The definition respects the ontological irreducibility of each layer while identifying their joint realization as a new primitive unit. The unit is not reducible to either Layer 1 or Layer 2 alone; its identity is relational-temporal.

14 Discussion and Open Questions

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1. How robust are episode detection and Joint Score to choice of embedding parameters, RQA configuration, and normalization conventions?
2. Can a fully continuous, non-thresholded characterization of the Joint Episode be developed (e.g., via integrated measures of permeability and intensification)?

3. What is the precise functional form relating Joint Score to Layer 3 transition probability across regimes?
4. How does the framework generalize to systems with continuous spatial structure or higher-dimensional attractors?
5. Ontological status of the "empty" Layer 2 interval (sustained low $A(k)$ without Layer 1).

16 Conclusions

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We have introduced the Joint Episode as a mathematically characterizable and ontologically significant unit. By grounding detection in sustained low antisynchrony containing Layer 1 intensification, defining the Joint Score, and demonstrating regime-dependent behavior in synthetic systems, the construct bridges recurrence quantification metrics and the dynamics of structural transition.

The ontological reading positions the Joint Episode as the operative temporal unit through which local persistence acquires the relational coherence required for global reorganization. Future work will extend calibration to additional real-world datasets and explore continuous formulations.

References